A SHEAF OF SETS OF BOUNDED SIZE ON THE SMALL ETALE SITE IS THE SHEAF OF SECTIONS OF AN ETALE SPACE OVER Y

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 03LP: sources/stacks-chow/tags.txt, source lines 4721–4721. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00WS: sources/stacks-chow/tags.txt, source lines 1194–1194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K8: sources/stacks-chow/tags.txt, source lines 5964–5964. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02WW: sources/stacks-chow/tags.txt, source lines 3819–3819. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03HU: sources/stacks-chow/tags.txt, source lines 4580–4580. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme and let Y be an algebraic space over S. Let $\mathcal{F}$ be a sheaf of sets on $Y_{étale}$. Provided a set theoretic condition is satisfied (see proof) we have*

1. *the functor X associated to $\mathcal{F}$ above is an algebraic space,*
2. *the map $X \to Y$ is an étale morphism of algebraic spaces,*
3. *via the identification $Sh(Y_{étale}) = Sh(Y_{spaces,étale})$ we have $\mathcal{F} \cong h_X$,*

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(4) *we have* $\mathcal{F} \cong f_{small,!}*$. *Here* $*$ *is the final object of the category* $Sh(X_{étale})$ *and* $f_{small,!}$ *exists by Lemma 03LP.*

Proof. Let us prove that X is a sheaf for the fppf topology. Namely, suppose that $\{g_i : T_i \to T\}$ is a covering of $(Sch/S)_{fppf}$ and $(y_i, s_i) \in X(T_i)$ satisfy the glueing condition, i.e., the restriction of (y_i, s_i) and (y_j, s_j) to $T_i \times_T T_j$ agree. Then since Y is a sheaf for the fppf topology, we see that the y_i give rise to a unique morphism $y : T \to Y$ such that $y_i = y \circ g_i$. Then we see that $y_{i,small}^{-1}\mathcal{F} = g_{i,small}^{-1} y_{small}^{-1}\mathcal{F}$. Hence the sections s_i glue uniquely to a section of $y_{small}^{-1}\mathcal{F}$ by Étale Cohomology, Lemma 09XN.

The construction that sends $\mathcal{F} \in \mathrm{Ob}(Sh(Y_{étale}))$ to $X \in \mathrm{Ob}((Sch/S)_{fppf})$ preserves finite limits and all colimits since each of the functors y_{small}^{-1} have this property. Of course, if $V \in \mathrm{Ob}(Y_{étale})$, then the construction sends the representable sheaf h_V on $Y_{étale}$ to the representable functor represented by V.

By Sites, Lemma 00WS we can find a set I, for each $i \in I$ an object V_i of $Y_{étale}$ and a surjective map of sheaves

$$\coprod h_{V_i} \longrightarrow \mathcal{F}$$

on $Y_{étale}$. The set theoretic condition we need is that the index set I is not too large[1]. Then $V = \coprod V_i$ is an object of $(Sch/S)_{fppf}$ and therefore an object of $Y_{étale}$ and we have a surjective map $h_V \to \mathcal{F}$.

Observe that the product of h_V with itself in $Sh(Y_{étale})$ is $h_{V \times_Y V}$. Consider the fibre product

$$h_V \times_{\mathcal{F}} h_V \subset h_{V \times_Y V}$$

There is an open subscheme R of $V \times_Y V$ such that $h_V \times_{\mathcal{F}} h_V = h_R$, see Lemma 04K8 (small detail omitted). By the Yoneda lemma we obtain two morphisms $s, t : R \to V$ in $Y_{étale}$ and we find a coequalizer diagram

$$h_R \rightrightarrows h_V \longrightarrow \mathcal{F}$$

in $Sh(Y_{étale})$. Of course the morphisms s, t are étale and define an étale equivalence relation $(t, s) : R \to V \times_S V$.

By the discussion in the preceding two paragraphs we find a coequalizer diagram

$$R \rightrightarrows V \longrightarrow X$$

in $(Sch/S)_{fppf}$. Thus $X = V/R$ is an algebraic space by Spaces, Theorem 02WW. This proves (1). Part (2) follows because $V \to Y$ is étale. Part (3) is immediate from the definition of X and h_X. We omit the proof of part (4); it follows by matching the morphism associated to the cocontinuous functor j of Lemma 03LP with the description of $X_{spaces,étale}$ as the localization of $Y_{spaces,étale}$ at X discussed above and Sites, Lemma 03HU. □

[1]It suffices if the supremum of the cardinalities of the stalks of $\mathcal{F}$ at geometric points of Y is bounded by the size of some object of $(Sch/S)_{fppf}$.